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MILITARY SPECIFICATION

COMPRESSORS, CENTRIFUGAL DISPLACEMENT TYPE, MOTOR DRIVEN, OIL-FREE AIR FOR NAVAL SHIPBOARD LOW PRESSURE AIR SERVICE

1. SCOPE

1.1 This specification covers stationary, centrifugal displacement (liquid piston), single stage class N, air compressors for shipboard service, driven by electric motors.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards and drawings, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

FF-B-171 - Bearing, Ball, Annular (General Purpose).
QQ-N-281 - Nickel-Copper-Alloy (Monel and R-Monel) Bars, Plates, Rods, Sheets, Strips, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (K-Monel).
QQ-N-288 - Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.
QQ-S-763 - Steel Bars, Shapes and Forgings - Corrosion Resisting.
QQ-S-766 - Steel Plates, Sheets and Strip - Corrosion Resisting.
GGG-P-781 - Puller and Puller Kit, Mechanical, and Mechanical Puller Attachment.

MILITARY

MIL-C-104 - Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116 - Preservation, Methods of.
MIL-R-196 - Repair Parts for Internal Combustion Engines, Packaging of.
JAN-P-735 - Primer, Paint, Zinc-Chromate, Alkyd Type.
MIL-G-858 - Gray Iron Alloy; Castings (Scale-Resisting).
MIL-S-890 - Steel: Forgings and Bars for Hulls, Engines, and Ordnance (Heat Treated).
MIL-S-901 - Shockproof Equipment, Class HI (High-Impact), Shipboard Application Tests for.
MIL-D-963 - Drawings, Production, Electrical Hull and Mechanical Equipment.
MIL-C-2174 - Controllers Motor Starters and Master Switches——Direct-Current, Naval Shipboard Use.
MIL-C-2212 - Controllers Motor Starters, and Master Switches - Alternating-Current, Naval Shipboard Use.
MIL-P-15024 - Plates, Identification———Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manual, Technical, for Mechanical and Electrical Equipment (Less Electronics).

FSC 4310

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MILITARY (cont'd.)

MIL-S-15083 - Steel Castings.
MIL-P-15130 - Paint, Outside, Haze-Gray, No. 27 (Formula No. 5H) (for Naval Shipboard Use).
MIL-P-15137 - Provisioning and Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-C-15730 - Coolers, Fluid, Industrial, Naval Shipboard: Lubricating Oil, Hydraulic Oil, and Fresh Water.
MIL-A-15939 - Aluminum Bronze Bars, Plates, Rods, Sheets, Strips, Forgings, and Structural and Special Shaped Sections.
MIL-S-16113 - Steel Plate, Hull and Ordnance, Structural, Black (Uncoated) and Zinc-Coated (Galvanized).
MIL-P-16298 - Preservation, Packaging, Packing, and Marking of Electric Machines Having Rotating Parts and Associated Repair Parts.
MIL-B-16444 - Bronze, Hydraulic (Ounce Metal): Castings.
MIL-B-16541 - Bronze, Valve: Castings.
MIL-M-16576 - Metal, Gun: Castings.
MIL-M-17060 - Motors; Alternating Current, Integral HP (Shipboard Use).
MIL-I-17166 - Iron Castings; Nodular Graphitic (Ductile Iron) (for Shipboard Applications).
MIL-I-17214 - Indicator, Permeability; Low-Mu (Go-No-Go).
MIL-M-17413 - Motors, Direct-Current, Integral HP, Naval Shipboard.
MIL-S-17509 - Steel Castings, Austenitic, Chromium-Nickel, Low Magnetic Permeability.
MIL-C-17516 - Copper-Silicon Alloy Bars, Plates, Rods, Sheets, Strips, Flat Wire, Wire, Forgings, and Structural and Special Shaped Sections.
MIL-P-17555 - Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts.
MIL-S-17996 - Steel Sheet and Other Wrought Forms, Corrosion Resisting Austenitic (Low Magnetic Permeability) Chromium-Manganese-Nickel.
MIL-G-18997 - Gages, Pressure, Dial Indicating, Bourdon Tube, Circular Face.
MIL-T-19646 - Thermometers, Remote Reading, Self-Indicating Dial, Gas Actuated.
MIL-S-20166 - Steel: Bars and Shapes (for Hull Construction) (Including Material for Drop and Miscellaneous Forgings).

NAVY DEPARTMENT

General Specifications for Inspection of Material.

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-278 - Welding and Allied Processes for Machinery for Ships of the United States Navy.

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DRAWINGS

BUREAU OF SHIPS
B-214 - Root Connections for Attaching Piping.
5000-S4823-F-841337 - Unions, Silver Alloy Brazing, for Oil, Water, Gas or Oxygen.
5000-S4800-841177 - Gage Connections.
810-1385850 - Piping Gage for Submarine Service.
810-1385875 - Plugs, Zinc, and Adaptors for Heat Exchangers.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

COMPRESSED AIR AND GAS INSTITUTE
Compressed Air Handbook.

(Application for copies should be addressed to the Compressed Air and Gas Institute, 122 West 42nd, New York, New York.)

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd St., New York 16, N. Y.)

NATIONAL BUREAU OF STANDARDS
Handbook H-28 - Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Qualification. - Compressors furnished under this specification shall be a product which has been tested and has passed the qualification tests specified herein (see 6.2).

3.2 Definitions. -

3.2.1 Actual free air capacity. - Actual free air capacity as used in this specification refers to the quantity of air, expressed in cubic feet per minute at intake conditions of temperature and pressure, which is actually delivered.

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3.3 General requirements. -

3.3.1 Assembly. - The equipment shall consist of a compressor and driving motor base mounted on a common horizontal bedplate, with separator, heat exchanger, electrical controls, and other items listed as follows, unless otherwise specified in the contract or order, and such other items as may be required for the satisfactory operation of the compressor:

Filter silencer, and air intake.
Gages, pressure.
Switch, pressure.
Valves, drain (air, water).
Thermometers.
Valve, check, air discharge.

A receiver will not be required unless otherwise specified (see 6.1). The compressor gage board, drain valves and all features of an operational nature shall be located on or shall be operable from one side of the compressor.

3.3.2 Basic requirements. - The principle of reliability is paramount and no compromise of this principle shall be made with any other basic requirements of design. It is the intention of this specification to obtain air compressors of such design that they will operate over a long period of years with a minimum of servicing. Where wear or corrosion is unavoidable the parts subjected to such wear or corrosion shall be of the best materials available for the purpose in order to reduce these detrimental effects to a minimum. The design and construction of all compressors shall be the most compact consistent with the following requirements:

(a) Reliability.
(b) Accessibility for repair.
(c) Resistance to wear or corrosion
(d) Economy.
(e) Satisfactory operation when inclined as follows:
    (1) Up to 15 degrees in any direction (permanently inclined).
    (2) With the ship rolling up to 60 degrees from the vertical.
    (3) With the ship pitching 30 degrees up and down from the normal horizontal plane.
(f) Minimum space and weight.

3.3.3 Maintenance. - The designer shall take cognizance of the conditions under which equipment will be maintained and repaired on shipboard, and of the fact that the personnel responsible for maintenance and repair may not be seasoned mechanics. For example, parts shall be so designed that it is impossible to replace them in a wrong position; adjacent studs shall be of the same length, or of large difference in lengths, so that a mistake will not easily occur; serrated surfaces shall not be used for accurate positioning, since the serrations are easily damaged; large wrench areas shall be avoided where strong wrenching would cause damage. All parts subject to wear or service failure shall be replaceable.

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3.3.4 Sizes. - Compressors shall be of one of the sizes shown in table I as specified (see 6.1).

Table I - Sizes.

Size	C.f.m.	Discharge pressure p.s.i.g.	Hp.
10A	10	85	5
15A	15	85	7-1/2
20A	20	85	10
50A	50	85	25
50B	50	100	25
100B	100	100	50

3.3.5 Standard product. - All equipment shall be essentially the product of the manufacturer wherever practicable so that prompt and continuing service and delivery of repair parts may be assured.

3.3.6 Design conditions. - Unless otherwise specified in the contract or order, the following design conditions shall be assumed:

(a) Ambient and intake air temperature - 122° Fahrenheit (F.).
(b) Intake air pressure - 14.7 pounds per square inch (p.s.i.) absolute.
(c) Relative humidity - 50 percent.
(d) Maximum permissible temperature of air leaving the compressor - 105°F.
(e) Temperature of cooling water at inlet - 85°F.
(f) Kind of cooling water - sea (salt).

3.3.7 Drive. - Direct connected motor drive is preferred although multi-"V" belt drive will also be considered. If belt driven, suitable means shall be provided for tightening and adjusting the belts. Protective guards shall be provided for all exposed moving parts.

3.3.8 Rating. - It shall be possible to operate each compressor in accordance with all the requirements of this specification at any pressure up to and including its maximum working pressure.

3.3.9 Shockproofness. -

3.3.9.1 All compressors shall be designed to withstand shock due to firing of the ship's own armament and noncontact underwater explosions of near-miss aerial bombs, torpedoes, and mines. Every attempt shall be made to design the equipment to resist shock as specified by graphs A, B, and C, shown on figure 1, it being understood that these shocks are applied at the footings of the component. Resilient mountings shall not be used unless bureau or agency concerned approval of the mount and its proposed application is given.

3.3.9.2 Holding-down bolts, supporting feet or legs, and subbases shall be designed to resist shock by using the accelerations specified on the graph shown on figure 1 for determining shock forces. The determination of stresses resulting from these forces is accomplished by the application of the usual methods of analytical mechanics. The allowable stress in components due to shock shall be the yield strength (0.2 percent offset) of the material. Normal stresses in parts, such as the tensile stress in a bolt due to initial pull-up, shall not be added to the shock stresses. Where two or more components are installed on a common rigid subbase, the feet and holding-down bolts, as well as the subbase, shall be designed for an acceleration based on the total weight of the assembly.

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3.3.9.3 The design of all complete compressor units shall be such that they are capable of passing the high-impact shock test specified in Specification MIL-S-901.

3.3.9.4 Bolts designed to be stressed in shear shall be installed in holes with a minimum of clearance, as experience indicates that large clearances allow impacting and subsequent failure.

3.3.10 Compressor details. -

3.3.10.1 Separator. - The separator shall be a tank type separator with suitable and accurate means for determining the level of the water in the tank while operating, and for checking the level for starting. The maximum and minimum allowable operating levels shall be clearly indicated. The baffles shall be designed for maximum effectiveness in removing entrained moisture from the discharge air. A drain cock shall be provided at the lowest point in the separator for lowering the water level, and for blow-off of impurities. The drain cock shall be readily accessible and visible from the operating side. A check valve shall be located in the air discharge line from the separator. The fresh water outlet shall be located at some point above the lowest point in the separator and shall lead direct to the heat exchanger.

3.3.10.2 Heat exchanger. - The heat exchanger shall be in accordance with Specification MIL-C-15730, except for submarine application (see 6.1), zinc pencil support plugs and vent and drain connection adapters shall conform to Drawing 810-1385875. Unless otherwise specified in the contract or order, the heat exchanger shall be of single tube sheet construction. Consideration shall be given in the design to the elimination of air pockets and the possibility of the heat exchanger becoming air-bound.

3.3.10.3 Noise. - The compressor shall be designed to operate as quietly as possible.

3.3.10.4 Alignment. - The case, frame, housing, and other fixed parts or attached components of the assembly shall be rigidly secured and held in alignment by dowel pins, fitted bolts, keys, rabbeted or tongue and groove joints, or other suitable means.

3.3.10.5 Special attention shall be given to the design of close-coupled compressors to insure that leakage from the stuffing box glands cannot enter motor frames or bearing brackets.

3.3.10.6 Bedplates. - The bedplates shall be continuous under the compressor and its driving unit. All top and bottom bearing surfaces shall be machine finished. Bedplates shall be sufficiently rigid to permit handling, shipment, and installation of the units onboard ship, without disturbing the alignment of the assembled units; and such that the normal distortion, weaving or vibration of the supporting structures onboard ships cannot cause misalignment of compressors.

3.3.10.7 Air filter. - Provisions shall be made for cleaning the inlet air and silencing the inlet air noise. These functions may be performed by a combination cleaner silencer, or separate devices for each function. Provision shall be made for replacement of cleaner elements which shall be the "dry" type. If necessary to achieve shock resistance of the unit, the inlet air connection shall consist of a slip joint nipple over which a length of flexible, spring wire reinforced ventilation hose may be fitted with clamps, the filter-muffler unit to be supported independently of the compressor. Not less than 5 feet of hose shall then be furnished as a part of the compressor.

3.3.10.8 Bearings. -

3.3.10.8.1 Radial and thrust bearings shall be of the ball type.

3.3.10.8.2 The use of ball bearings shall be subject to the speed limitations of Specification FF-B-171, except where prior approval by the bureau or agency concerned has been granted.

3.3.10.8.3 Bearings shall be installed in housings separate from and independent of the stuffing boxes and with adequate protection from gland leakage. Suitable means shall be provided to prevent escape of lubricant around the shaft. All baffles, wipers, and related parts shall be readily renewable.

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3.3.10.8.4 Lubrication of ball bearings shall be in accordance with Specification FF-B-171.

3.3.10.9 Piping and valves. - All water and air connections to the compressor shall be readily accessible. Pockets where water may collect shall be avoided, or if unavoidable, shall be arranged for convenient drainage. All compressor water connections to external shipboard piping systems shall be flanged. Flange and bolting standards shall be in accordance with Drawing B-214. Air root connections shall be in accordance with Drawing 5000-S4823-F-841337. All air and water piping on the unit shall be made up with connections so as to be readily removable for overhaul of the unit. Threaded pipe connections shall be reduced to an absolute minimum on internal piping. Compression type or flanged fittings shall be used wherever possible.

3.3.10.10 Threaded parts. - All threaded parts shall be in accordance with Handbook H-28. Use of studs shall be avoided.

3.3.10.11 Fresh water connections. - Provision shall be made for permanently connecting the unit to the ship's fresh water system through a spring loaded supply valve and a check valve. A strainer shall be installed in the fresh water line.

3.3.10.12 Salinity measurements. - A simple means of measuring the salinity of the fresh water shall be provided. The sensitivity of a laboratory type of sodium chloride tester shall be considered adequate. The indicator may be portable in which case it shall be considered a special tool (see 3.6.3). The manufacturer shall specify all salinity tolerances (warning plate and technical manual data) which shall be below those conducive to crystal formation which will affect the compressor unit or foul the air system.

3.3.10.13 Cooling water system. - All parts of the cooling system shall be of corrosion resisting material and shall be designed for the use of salt water supplied from the ship's flushing or fire mains, or when specified (see 6.1), from an attached or independent circulating water pump. Unless otherwise specified in the contract or order, the cooling system shall be designed for a working pressure of 50 p.s.i. on the cooling water inlet. Provision shall be made for completely draining all parts of the circulating water system.

3.3.10.13.1 Provision shall be made for positive and automatic control of the cooling water so that the water is turned on when the compressor is running and turned off when the compressor is not running (see 3.5.2.1.4). Automatic valves shall be provided with a means of manually opening or by-passing the valves in case of failure of the automatic feature.

3.3.10.14 Gages and gage board. - All gages shall be class a as specified in Specification MIL-G-18997. All gages and instruments shall be mounted on a common gage board. The gage board shall be provided with brackets for mounting on the compressor. Mounting brackets shall be of a type which can be removed from the compressor when the gage board is bulkhead mounted. Pressure gages shall be provided to indicate the air-delivery pressure and the pressure in the cooling water system. All gages on the gage board shall be suitably identified on an identification plate located on the gage board adjacent to each gage. The gages shall be marked to show the normal designed working pressure, and such working pressures shall be indicated on drawings and in technical manuals. Drawings 5000-S4800-841177 and 810-1385850 shall be used as a guide for gage connections.

3.3.10.15 Thermometers. - Thermometers shall be provided for the measurement of the temperature of the final discharge air and the fresh water at the compressor inlet. The thermometers shall be 4-1/2 inch dial, remote reading, vapor or gas actuated.

3.3.10.15.1 Vapor. - Vapor thermometers shall be in accordance with Specification MIL-T-15942 except as follows:

(a) Vapor actuated.
(b) Plastic lens.

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3.3.10.15.2 Gas. - Gas actuated thermometers shall be in accordance with Specification MIL-T-19646.

3.3.11 Couplings. -

3.3.11.1 Flexible couplings, when required, shall be of all metal construction.

3.3.11.2 All flexible coupling hubs shall be keyed to the shafts and secured by locknuts.

3.3.12 Welding. - Welding and allied processes shall be in accordance with Standard MIL-STD-278.

3.3.13 Painting. - All surfaces to be painted shall be thoroughly cleaned and free from oil, grease, mill scale, rust and loose weld beads. All external nonworking ferrous and aluminum surfaces shall be painted with one coat of zinc chromate primer in accordance with Specification JAN-P-735, and two coats of haze-gray paint in accordance with Specification MIL-P-15130.

3.3.14 Identification plates. -

3.3.14.1 Identification plates shall be furnished on each compressor and shall be type A or B in accordance with Specification MIL-P-15024, except that identification plates of plastic or aluminum are not acceptable.

3.3.14.2 Identification plates shall be secured to equipment with corrosion resistant metallic screws.

3.3.14.3 Identification plates shall contain data as follows:

(a) Manufacturer's name.
(b) Manufacturer's model or type and size.
(c) Classification.
(d) Manufacturer's serial number.
(e) Salient design characteristics:
    (1) Capacity.
    (2) Pressure in pounds per square inch.
    (3) Speed of shaft in r.p.m.
    (4) Brake horsepower.
    (5) Cooling water pressure drop at rated flow.
    (6) Any special data vital to the unit.
(f) Contract number (and item number of multiple unit orders).
(g) Stock number (space for number should be provided even though number may not be assigned).
(h) Section for inspector's stamp.

3.3.14.4 A view of the compressor identification plate shall be shown on the outline or master drawing showing all available data listed in 3.3.14.3, except that if shown on the master drawing serial number, contract number, and stock number data shall be omitted.

3.3.14.5 Each driving unit and each accessory unit shall have an identification plate in accordance with the applicable equipment specification.

3.4 Materials. -

3.4.1 General materials. -

3.4.1.1 All material used in the construction of compressors and associated equipment shall be of the best quality for the purpose intended.

3.4.1.2 The use of materials differing in properties from those specified for a given application will be considered when the contractor shows the necessity for such substitution, the material proposed is readily obtainable, and provided that the material is satisfactory to the bureau or agency concerned.

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3.4.1.3 Where no specifications are specified herein for the material for certain parts, the material used shall be of a type and quality consistent with that specified for associated parts in the compressor. The use or substitution of manufacturer's material shall be subject to approval by the bureau or agency concerned and shall be clearly indicated in a list submitted with the master drawings.

3.4.2 Specific materials. - Material used in the construction of compressors and associated equipment shall be identical to or the commercial equivalent of materials indicated in table II. In general, cast iron will not be permitted except for such items as nonstructural parts where experience indicates the use of cast iron is satisfactory. Cast steel or nodular cast iron shall be used in lieu of cast iron.

Table II - Material.

Application	Material	Specification
Bedplates	Cast steel	MIL-S-15083 or MIL-S-17509
	Structural steel, welded	MIL-S-20166; MIL-S-16113
	Corrosion resisting steel	QQ-S-763; QQ-S-766 or MIL-S-17996
	Aluminum bronze	MIL-A-15939
Casings Mounting brackets for close-coupled units	Gun metal	MIL-M-16576
Gland	Valve bronze	MIL-B-16541
Rotor	Gun metal Valve bronze	MIL-M-16576 MIL-B-16541
Impeller	Ni-Cu alloy	QQ-N-288 or QQ-N-281
Cone Lobe Casing cover	Gun metal Valve bronze Hydraulic bronze	MIL-M-16576 MIL-B-16541 MIL-B-16444
Shaft	Forged steel, class A or B Ni-Cu alloy, class a	MIL-S-890 QQ-N-281
Shaft sleeves	Ni-Cu-Al alloy, minimum hardness 265 Brinell	QQ-N-286
Separator	Copper-silicon alloy Ni-Cu alloy	MIL-C-17516 QQ-N-281
Sheaves	Cast steel	MIL-S-15083 or MIL-S-17509
	Cast iron	MIL-G-858
	Valve bronze	MIL-B-16541
	Nodular graphitic iron	MIL-I-17166

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3.4.3 Nonmagnetic construction. -

3.4.3.1 When the contract or order specifies that the compressor units shall be nonmagnetic the materials of construction shall be selected such that the magnetic permeability of the compressor component unit (less motor and controller) after fabrication will be 2.0 or less measured by a permeability indicator conforming to Specification MIL-I-17214.

3.4.3.2 In order to achieve the desired low permeability after fabrication it is advisable to choose materials which have a permeability prior to fabrication of 1.2 or less when materials whose permeability is affected by fabrication (CRES) are used.

3.5 Electrical equipment. -

3.5.1 Motors. - All motors shall be in accordance with Specification MIL-M-17413 for direct current (d.c.) motors, or Specification MIL-M-17060 for alternating current (a.c.) motors, and the following:

Service	Navy A.
Voltage and current characteristics	As specified (see 6.1).
Enclosure	Dripproof protected.
Ambient temperature	50°C.
Duty	Continuous.
Speed	Constant (see 3.5.1.2).
Winding	A.c., squirrel cage induction. D.c., stabilized shunt (see 3.5.1.2).
Insulation	Class A or B.
Bearings	Ball.

3.5.1.1 The horsepower rating of each motor shall be in accordance with the Navy standard motor ratings, but not less than 1.05 times the brake horsepower of the driven compressor under rated conditions of capacity and pressure.

3.5.1.2 Submarine d.c. motor applications. - D.c. motors for submarine application shall be suitable for the varying voltage specified in the contract or order. These motors shall be compound wound and shall have adjustable varying speed characteristics.

3.5.1.2.1 The hot full-field no-load speed at any voltage from 70 percent of nominal to 142 percent of nominal, shall not exceed 90 percent of the maximum safe operating speed of the compressors as established by the compressor manufacturer. These speeds shall be shown on the motor and compressor drawings.

3.5.1.2.2 Motors shall be capable of being started at all voltages within the varying voltage range specified, assuming that at the instant of starting the compressors will always be unloaded.

3.5.1.2.3 Motors shall be equipped with overspeed trips unless the conditions specified herein are met. Motors shall be mechanically capable of operating at a speed of 10 percent above the setting of the overspeed trip. If no overspeed trip is provided then the motors shall be mechanically capable of operating at a speed of 10 percent above the hot no-load speed at 142 percent rated voltage and with a field circuit resistance equal to 110 percent of the rated resistance value of the field rheostat. This speed shall be shown on the motor drawings.

3.5.2 Controllers and master switches. - One controller and associated master switches shall be supplied with each compressor. Controllers and master switches shall be in accordance with Specification MIL-C-2212 for a.c. equipment, or Specification MIL-C-2174 for d.c. equipment and the following:

Enclosure	Dripproof.
Operation	Magnetic.
Ambient temperature	50°C.
Performance	See 3.5.2.1.

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Control function	Motor starting.
Type	A. c. across line. D. c. resistor, inductive time lag acceleration.
Protection	Overload, low voltage (l. v. p.).
Duty	Continuous.
Master switches required	See 3. 5. 2. 1.
Master switch enclosures	Spraytight (remote) dripproof (local).

3. 5. 2. 1 Except for d. c. submarine service each controller shall be provided with local "ON, OFF", push buttons, a two-position selector switch marked "START-STOP" and "CONSTANT SPEED", and a pressure switch. The push buttons shall be of the momentary contact type.

3. 5. 2. 1. 1 With the selector switch in the START-STOP position, pressing the "ON" push button shall place the compressor in operation under automatic control of the pressure switch. When the air pressure in the discharge line is below the value set on the pressure switch, the pressure switch shall close and start the compressor. The compressor shall continue to run until the pressure rises to the value required to open the pressure switch causing the controller to stop the compressor. This start-stop operation shall continue in effect until the "OFF" push button is pressed.

3. 5. 2. 1. 2 With the selector switch in the "CONSTANT SPEED" position the compressor shall start when the "ON" push button is pressed (regardless of air pressure in the discharge line) and shall continue to run until the "OFF" push button is pressed. Under this control the compressor discharge pressure shall be kept below a set value by a pressure operated control valve.

3. 5. 2. 1. 3 The pressure switch shall be in accordance with Specification MIL-C-2174 or MIL-C-2212, as applicable. The pressure switch shall be of an adjustable type, suitable for the maximum pressure specified for the compressor and with a maximum differential of 30 p. s. i. or more. Unless otherwise specified in the contract or order, the pressure switch differential shall be not less than 20 p. s. i.

3. 5. 2. 1. 4 Cooling system valves (see 3. 3. 10. 13. 1) if electrically operated, and unloading devices if electrically operated shall conform to all the applicable requirements for controllers as specified in 3. 5. 2, except that enclosures shall be watertight (or better). Valves shall be tight-seated and all internal parts in contact with water shall be nonferrous material except where ferrous parts are necessary for the magnetic circuit, and such ferrous parts shall be suitably plated for protection against corrosion.

3. 5. 2. 2 Submarine d. c. controller applications. - D. c. controllers for submarine application shall be suitable for starting and controlling the motors at all voltages within the varying voltage range specified (see 3. 5. 1. 2).

3. 5. 2. 2. 1 A field accelerating relay shall be provided only if the motor manufacturer considers such a device essential for this application. If a field accelerating relay is provided, the voltage range over which the motor is required to start shall be fully considered in designing the relay coils.

3. 5. 2. 2. 2 Special attention shall be given to keeping weight and size to a minimum. To this end the number of accelerating steps shall be kept to a minimum. Stock sizes of controller enclosures which are larger than necessary for the equipment shall not be used.

3. 5. 2. 2. 3 Provision shall be made in the control circuits for connections to overspeed devices (see 3. 5. 1. 2. 3).

3. 6 Repair parts and tools. -

3. 6. 1 Repair parts and tools shall be furnished and processed in accordance with Specification MIL-P-15137.

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3.6.2 Onboard repair parts. -

3.6.2.1 Each set of repair parts required (see 6.1) shall contain such parts as listed in table III.

Table III - Onboard repair parts.

Name of part	Quantity per set
Ball or roller bearings Internal water lubricated bearings Coupling bushings, pins and other wearing parts Packing and gaskets Mechanical seals Shims	100 percent complete replacement for all installed units per ship.
Sleeve bearings Gland Lantern rings (seal cages) Stuffing box throat bushings Automatic valve diaphragms Automatic valve springs Fresh water valve springs	One to 4 units per ship: 100 percent replacement for one unit Over 4 units per ship: 100 percent replacement for two units
Cone Lobe Rotor, completely assembled, including shaft, impeller, wearing rings, diaphragms, bushings, sleeves, keys, flinger, nut, and complete coupling	100 percent replacement for one pump

3.6.2.2 Compressor rotor assemblies for motor driven close-coupled units shall be furnished disassembled, and shall include a complete motor rotor.

3.6.2.3 The compressor rotor assembly for flexible coupled units shall be completely assembled ready for installation in the compressor, and shall include any nonrotating parts which can be installed only by removing some rotating part or parts of the compressor rotor. The coupling or sheave shall be included.

3.6.2.4 Repair parts for heat exchangers, motors, controllers, pressure switches and other devices shall be furnished as required by the applicable specifications.

3.6.3 Special tools. -

3.6.3.1 Maintenance tools. - Each set of special maintenance tools required (see 6.1) shall consist of such tools as may be necessary for the proper repair and maintenance of the compressors. Each set shall be furnished in a separate steel box.

3.6.3.1.1 Note to ordering activities. - Normally the number of sets of special maintenance tools should be the same as the number of sets of onboard repair parts. However the number of sets of special tools may be increased as required by the size and organization of the ships.

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3.6.3.2 Operating tools. - Each set of special operating tools required by a contract or order shall consist of tools normally required to operate one compressor. Each set shall be furnished in a separate steel box.

3.6.3.2.1 Note to ordering activities. - Normally one set of special operating tools shall be ordered for each compartment in which an air compressor is installed.

3.6.3.3 Tools which are available in the Navy Stock List of General Stores, (copies of this stock list may be consulted in the office of the Government inspector), such as common wrenches and standard pullers, will not be required. Standard complete or limited sets of pullers or parts of puller sets in accordance with Specification GGG-P-781 shall be identified as to type and use in notes on the outline or section assembly drawings. Special tools shall be detailed, and included in the list of material.

3.6.4 Stock. - The selection, stock numbering and quantities of repair parts for stock shall be in accordance with Specification MIL-P-15137.

3.6.4.1 Unless otherwise specified in the contract or order, the cognizant supply demand control point for repair parts for stock is Ships Parts Control Center, Mechanicsburg, Pennsylvania.

3.7 Drawings. -

3.7.1 Drawings shall be type I in accordance with Specification MIL-D-963.

3.7.2 A separate outline drawing shall be furnished for each contract or order and for each different compressor unit on a contract or order. The outline drawing shall apply to the complete compressor assembly, including compressor, driver, bedplate and attached accessories. In addition to the data specified in Specification MIL-D-963, the center of gravity for the compressor assembly, wet, shall be shown on this drawing.

3.7.2.1 If the compressors and motors are close-coupled or built together so that they are not in fact separate components then a master drawing and certification data sheet or sheets may be substituted for the outline drawing. With this arrangement the master drawing shall contain everything required for the outline drawing except certification data applying to an individual contract or order which shall be included on the compressor certification data sheet or sheets. Unless otherwise specified in the contract or order, separate certification data sheets will also be required for the motor, the controller and heat exchanger.

3.7.3 Compressor master drawings shall include sectional assembly, and elevation drawings, and details of all parts. The sectional assembly and elevation drawings shall identify all parts and shall include a complete list of material with references to the detail drawings. The details shall be on one or more additional sheets. Size F drawings (28 by 40 inches) are preferred for sectional assembly and detail drawings. In any event drawings shall not exceed 84 inches in length.

3.7.3.1 If more than one sectional assembly or elevation drawing is required to identify all parts, or if the sectional assembly with list of material results in a drawing larger than 28 by 50 inches, then the list of material may be on sheets separate from but bearing the same drawing numbers as the sectional assembly.

3.7.3.2 It is recommended that details of parts common to all sizes of compressors be grouped together on the same sheet(s); and likewise all details applying to a single size should be grouped together on the same sheet(s).

3.7.3.3 The sectional assembly drawing shall show all assembly clearances, and shall reference the qualification test, high impact shock test, and inclination test reports, when such reports are required, and the Bureau of Ships letters approving those reports.

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3.7.4 Subassemblies furnished as repair parts in lieu of or in addition to individual repair parts shall be recorded on the related outline drawings by reference notes or as additional items on the "List of Material" of the sectional assembly drawing.

3.7.5 Drawings for drivers and associated equipment shall be in accordance with the specifications for those equipments.

3.7.6 Drawings submitted with bids are not required to be in master drawing format.

3.8 Technical manuals. - Unless otherwise specified in the contract or order, technical manuals shall be in accordance with type III of Specification MIL-M-15071. A separate manual shall be furnished for each different compressor unit except in special cases when manuals covering more than one compressor may be approved. Preliminary manuals submitted for approval shall include all proposed sections complete.

3.8.1 Illustrations. - Two photographs, taken 180 degrees apart on a horizontal plane shall be included. Reduced size reproductions of approved outline, assembly and sectional drawings of the complete unit shall be included. Reduced size reproductions of approved electrical drawings and certification data shall also be included. Approved drawing excerpts and exploded views of parts or assemblies requiring detailed instruction shall be included.

3.8.2 Distribution. - Two copies of technical manuals shall be packed with each compressor unit. Additional copies of manuals shall be furnished as specified (see 6.1). If final manuals are not available at the time compressors are shipped then two copies of preliminary manuals shall be packed with each compressor unit; and the final manuals shall be delivered at a later time in accordance with Specification MIL-M-15071.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Sampling. -

4.2.1 Lot. - A lot shall consist of a group of identical compressors produced on one contract or order at the same manufacturing plant.

4.2.2 Sampling for lot acceptance tests. - Completely assembled compressors, including driving units, and all equipment or fittings shall be selected by the Government inspector in accordance with table IV for the tests specified in 4.3.4.

Table IV - Sampling.

Number of identical units on order	Random selection
Up to 30	1 from every 10 units
31 to 100	1 from every 20 units, but not less than 4
Above 100	1 from every 20 units, but not less than 10

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4.3 Tests.-

4.3.1 All tests specified herein shall be at the expense of the exhibitor unless otherwise specified or approved by the bureau or agency concerned.

4.3.2 Test reports.- Test reports, as specified in 4.3.11, shall be prepared for all tests conducted by the contractor or laboratory, including the shock and inclination tests.

4.3.3 Hydrostatic test.- Each compressor shall be hydrostatically tested at the place of manufacture in accordance with the following:

(a) Compressor casing, separator and cooling water side of heat exchanger, -150 p.s.i.g.
(b) Other parts subjected to pressure shall be tested to 1-1/2 times the maximum designed working pressure.

4.3.4 Lot acceptance tests.- Compressors selected in accordance with 4.2.2 shall be tested at the place of manufacture in accordance with 4.3.7 and 4.3.8.

4.3.5 The remaining compressors shall be subjected to the tests specified in 4.3.8.2.

4.3.6 Qualification tests.- Qualification tests shall be conducted at the place of manufacture for the hydrostatic test specified in 4.3.3 and at a laboratory satisfactory to the Bureau of Ships for the tests specified in 4.3.7 to 4.3.10, inclusive. Qualification tests shall be conducted on each size to determine that performance and design of the compressor are in compliance with this specification.

4.3.7 Component tests.- Motors, controllers and heat exchangers shall be tested as required by the applicable specifications.

4.3.8 Performance tests.-

4.3.8.1 Compressor units shall be submitted to a continuous running test of 8 hours' duration at normal pressure. Test is to be run under normal electrical conditions specified (see 6.1). Power consumption tests shall be conducted as a part of the tests. In these tests the following data shall be recorded at the start and finish of the run and at regular intervals, not exceeding 1/2 hour throughout the run.

4.3.8.1.1 Capacity test.- The measurement of delivered air in accordance with the A.S.M.E. Long-Radius Low-Ratio nozzle method specified in the Compressed Air Handbook is preferred. The multiple orifice method may be used if it is the manufacturer's standard method of capacity measurement.

4.3.8.1.2 Recorded data.-

MECHANICAL

(a) Dry bulb ambient temperature.
(b) Wet bulb ambient temperature.
(c) Intake temperature (in air stream close to intake).
(d) Barometer readings.
(e) Air discharge pressure.
(f) Cooling water inlet and outlet temperatures.
(g) Cooling water inlet and outlet pressures.
(h) Quantity of cooling water in gallons per minute.
(i) Revolutions per minute of driving unit and of compressor.

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ELECTRICAL

(a) Voltage, amperage and frequency.
(b) Power factor.
(c) Power input to motor.
(d) Complete data as to performance of motor including all data required to determine suitability of the electrical equipment for the specific application. This data shall be recorded primarily to demonstrate that the motor is of adequate power for the service intended.

OPERATIONAL

One of the compressors selected in accordance with 4.2 shall be given additional tests to provide sufficient data to prepare characteristic curves as follows:
(a) Capacity versus discharge pressure.
(b) Brake horsepower versus discharge pressure.
The curves shall be plotted and shall cover at least that portion of the performance from 60 p.s.i.g. to 90 p.s.i.g. discharge pressure for 85 p.s.i. units, and 60 p.s.i.g. to 105 p.s.i.g. for 100 p.s.i. units.

4.3.8.1.3 Computed data. - The following data shall be computed from the recorded data and shall be included on each test data record:

(a) Capacity in cubic feet of free air per minute actually delivered, computed as indicated in the Compressed Air Handbook. If a test method other than the A.S.M.E. Long-Radius Low-Ratio nozzle method is used, then each set of test reports shall include a description of the method of computations used, and a sample of actual computations applying to at least one of the 8-hour tests.
(b) Brake horsepower at the compressor shaft (computed from previous test of motor).
(c) Overall efficiency of compressor. Theoretical air horsepower upon which efficiency is based may be calculated from either the adiabatic or isothermal base, but test data should indicate which is used.

4.3.8.1.4 During each test the functioning of the compressor units shall be smooth and free from excessive heating, noise, and vibration and the performance in all respects shall meet the requirements of this specification.

4.3.8.1.5 Upon completion of the performance tests, each compressor shall be disassembled sufficiently to permit examination of all parts subject to wear or damage by running. To be acceptable no parts of the compressors shall show any signs of failure or excessive wear. The condition of the compressors at the conclusion of all tests shall be noted on the official test sheets.

4.3.8.2 Each compressor unit not selected in accordance with 4.2.2 shall be subjected to tests exactly as required by 4.3.8.1 except:

(a) The duration of the test need not exceed 4 hours, and data is required to be taken only at the start and finish of the run and once in between.
(b) The units need not be disassembled upon completion of the performance tests, unless excessive noise, vibration, or erratic performance is noted during the tests.

4.3.8.2.1 If excessive noise, vibration, or erratic performance is noted and is corrected, the 4-hour test shall be repeated. If the first correction does not completely eliminate the trouble, then after further correction the compressor shall be subjected to a complete 8-hour test before acceptance.

4.3.8.2.2 All recorded data and computation required for the 8-hour test are also required for the 4-hour test.

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4.3.9 Inclination test. -

4.3.9.1 One compressor of each design including the driving motor shall be subjected to the inclination tests to determine satisfactory compliance with 3.3.2 and the tests need not be repeated unless the Government inspector or bureau or agency concerned considers that sufficient changes have been made from the tested design to warrant retesting. Neither changes in materials only of compressor construction, nor changes in the driving motor shall be considered as reasons for retesting.

4.3.9.2 The compressor unit shall be operated at rated load for not less than 30 minutes with the base tilted 15 degrees in each of the four cardinal positions, right, left, front, and rear. During the progress of these tests there shall be no excessive heating of any parts, and noise and vibration shall not exceed that of normal horizontal operation.

4.3.9.3 The suitability of the compressor unit for operation when a ship is rolling or pitching shall be determined by examination of the design.

4.3.10 High-impact shock test. -

4.3.10.1 One complete compressor unit of each design and size shall be selected by the Government inspector for type A shock tests in accordance with Specification MIL-S-901.

4.3.10.1.1 Only one compressor of each type, design, and size shall be subjected to the shock tests, and the tests need not be repeated on later contracts or orders unless the Government inspector or bureau or agency concerned considers that sufficient changes have been made from the tested design to warrant retesting. Changes in materials of construction may require retesting. Changes in the driving motor only shall not be considered a reason for retesting a compressor; however changes in the motor will require shock testing of the motor.

4.3.10.1.2 If a manufacturer is unable to conduct the shock tests at his own plant, he may arrange to have them conducted at a commercial or Government laboratory suitably equipped to conduct the tests. When the shock tests are to be conducted at a Government laboratory a complete set of master and outline drawings shall accompany the equipment.

4.3.10.2 The compressor shall be idle and running on alternate blows of the shock test, but the compressor shall not be compressing air while running.

4.3.10.3 Examination after shock test. -

4.3.10.3.1 Motor. - Examination of the motor shall be in accordance with Specification MIL-M-17060 or MIL-M-17413.

4.3.10.3.2 Compressor. - Examination of the compressor shall be conducted as follows:

(a) Examine the unit carefully and note the extent of any external damage.
(b) Turn the compressor over several times by hand to insure that all working parts are free and ready for starting.
(c) Run the compressor unloaded for a short time and note any mechanical pecularities.
(d) Gradually build up the load and carefully check any undue noises or excessive vibration or other nonconformances.
(e) Secure the compressor and disassemble for complete examination. Observe and record all effects of the shock test on the various compressor components.

4.3.10.4 Failure to pass test. -

4.3.10.4.1 Motor. - The motor will be considered to have failed the shock test unless it conforms to Specification MIL-M-17060 or MIL-M-17413.

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4.3.10.4.2 Compressor. - The compressor will be considered to have failed the shock test in the event of the following:

(a) Breakage of any essential parts, including mounting bolts.
(b) Detachment of any part or object from the unit.
(c) Compressor requires attention or adjustment or replacement of parts to enable it to perform its principal function during and after each blow of the test.
(d) Unit fails to function satisfactorily as specifically defined in this specification.

4.3.10.5 Disposition of shock-tested equipment. - Equipment which has successfully passed the shock tests will be considered acceptable for such service as designated by the bureau or agency concerned. The contractor shall thoroughly recondition such units, correcting any minor imperfections or deformations, and shall place the equipment in satisfactory working order. The motors shall be reconditioned and tested in accordance with the requirements of the motor specifications. After reconditioning, the equipment shall be tested as specified in 4.3.8.2.

4.3.10.5.1 Equipment which has been subjected to the high-impact shock tests and has failed will not be accepted, either in whole or in part until the causes of such failure have been corrected by the manufacturer, and the complete equipment has been successfully retested. However, failure of the compressor only shall not cause rejection of an otherwise acceptable motor, nor shall failure of a motor only cause rejection of an otherwise acceptable compressor.

4.3.11 Test reports. -

4.3.11.1 Test reports shall be prepared for all tests conducted by the contractor or laboratory, including the shock and inclination tests. Recorded data of the compressor performance tests may be logged on contractor's forms. The following identification data for each compressor unit shall be furnished and included in the test report:

(a) Serial numbers of the compressor and driving unit.
(b) Ship for which intended.
(c) Contract or order number.
(d) Actual net weight of the unit in the "dry" and "wet" condition.

4.3.11.2 Distribution of compressor performance test reports shall be as follows:

(a) Two to the bureau or agency concerned.
(b) Two to each cognizant supervisor of shipbuilding for each ship's worth of compressors furnished (one for Supervisor's files; balance for shipbuilder) for privately built ships.
(c) Two to each cognizant Naval Shipyard for each ship's worth of compressors for Navy built ships.
(d) One for primary inspection officer.
(e) One for branch inspection office (if any).

4.3.11.3 Distribution of shock test reports and inclination test reports shall be as follows:

(a) Six to the bureau or agency concerned, for approval.
(b) One for the primary inspection office.
(c) One for the branch inspection office (if any).

4.4 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging (see 6.1). -

5.1.1 Level A. - Motor driven compressors shall be preserved and packaged in accordance with method IIa of Specification MIL-P-116. Contact preservatives shall not be required. Heat exchangers and separators shall be dried by passing dry air through the system. Inspection ports or other openings to the heat exchanger, separators or compressor shall be left open to permit maximum circulation of air. Detached controllers and electrical components shall be preserved and packaged in accordance with Specification MIL-C-2212, MIL-P-16298 or MIL-P-17555, as applicable.

5.1.1.1 Repair parts shall be preserved and packaged in accordance with Specifications MIL-P-17555 and MIL-R-196 as applicable.

5.1.2 Level C. - Preservation and packaging of compressors shall be in accordance with the manufacturer's commercial practice.

5.2 Packing (see 6.1). -

5.2.1 Levels A and B - compressors and detached components. - Compressors and detached components, preserved and packaged by level A or C, shall be packed in accordance with Specification MIL-P-17555. An inspection door, in accordance with Specification MIL-C-104, shall be provided in the shipping crate and shall be positioned to permit inspection of the humidity indicator in the method IIa package. Detached components shall be packed within the same shipping container with the compressor.

5.2.2 Level C. - Compressors and detached components, packaged as specified, shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply with the Uniform Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.2.3 Repair parts. - Repair parts, preserved and packaged as specified, shall be packed for the level specified in accordance with Specification MIL-P-17555.

5.3 Marking. - In addition to any special marking required by the contract or order or herein, interior and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether receiver is required (see 3.3.1).
(c) Floor space, height and weight limitations, if any. Limits of height, length, and width for disassembly, if any (see 3.3.2(f)).
(d) Size of compressors (see 3.3.4).
(e) Operating conditions, under which the compressor shall be capable of operating satisfactorily (see 3.3.6 and 4.3.8.1).
(f) Whether for submarine application (see 3.3.10.2).
(g) Whether circulating water pump is required (see 3.3.10.13).
(h) Characteristics of electrical supply (see 3.5).
(i) Repair parts requirements (see 3.6).
(j) Tool requirements (see 3.6.3).
(k) Number of technical manuals required (see 3.8.2).
(ℓ) Level of preservation, packaging, and packing (see 5.1 and 5.2).

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6.2 With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening of bids, been tested and approved for inclusion in Qualified Products List QPL-19553, whether or not such products have actually been so listed by that date.

6.2.1 The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government, tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Chief of the Bureau of Ships, Department of the Navy, Washington 25, D. C., and information pertaining to qualification of products may be obtained from that activity.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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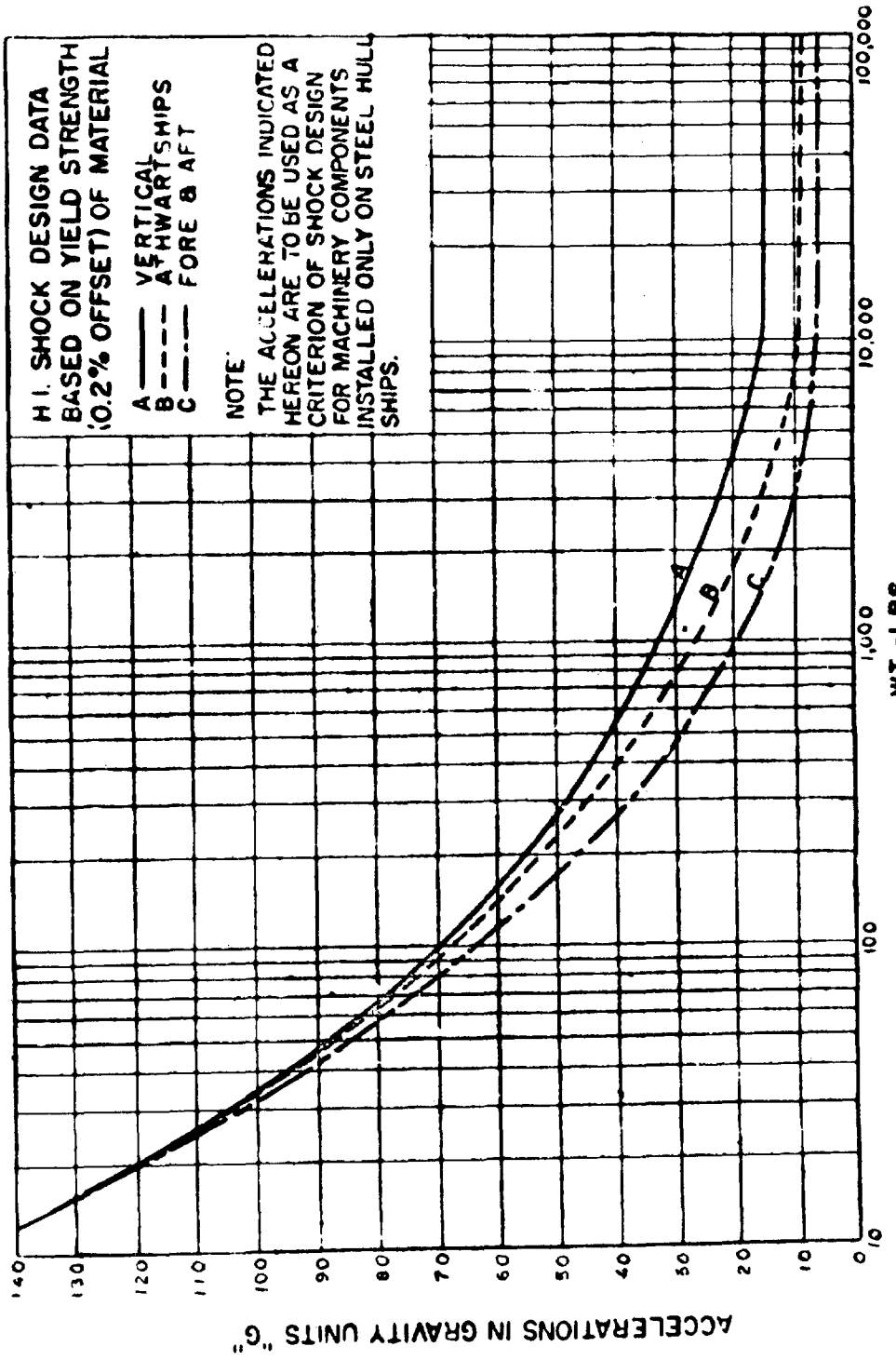


Figure 1.-H.I. Shock design data.

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